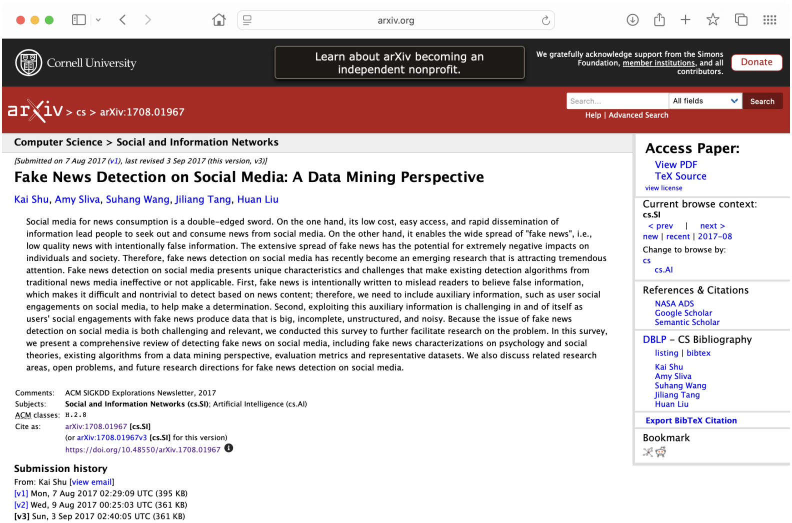
Decision and Risk Analysis

Lecture 2: **Bayes' Rule**

DAEN 427/ISEN 427/ISEN 627

Alexander Abuabara

Fall 2026



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[Submitted on 7 Aug 2017 (v1), last revised 3 Sep 2017 (this version, v3)]

Fake News Detection on Social Media: A Data Mining Perspective

Kai Shu, Amy Sliva, Suhang Wang, Jiliang Tang, Huan Liu

Social media for news consumption is a double-edged sword. On the one hand, its low cost, easy access, and rapid dissemination of information lead people to seek out and consume news from social media. On the other hand, it enables the wide spread of "fake news", i.e., low quality news with intentionally false information. The extensive spread of fake news has the potential for extremely negative impacts on individuals and society. Therefore, fake news detection on social media has recently become an emerging research that is attracting tremendous attention. Fake news detection on social media presents unique characteristics and challenges that make existing detection algorithms from traditional news media ineffective or not applicable. First, fake news is intentionally written to mislead readers to believe false information, which makes it difficult and nontrivial to detect based on news content; therefore, we need to include auxiliary information, such as user social engagements on social media, to help make a determination. Second, exploiting this auxiliary information is challenging in and of itself as users' social engagements with fake news produce data that is big, incomplete, unstructured, and noisy. Because the issue of fake news detection on social media is both challenging and relevant, we conducted this survey to further facilitate research on the problem. In this survey, we present a comprehensive review of detecting fake news on social media, including fake news characterizations on psychology and social theories, existing algorithms from a data mining perspective, evaluation metrics and representative datasets. We also discuss related research areas, open problems, and future research directions for fake news detection on social media.

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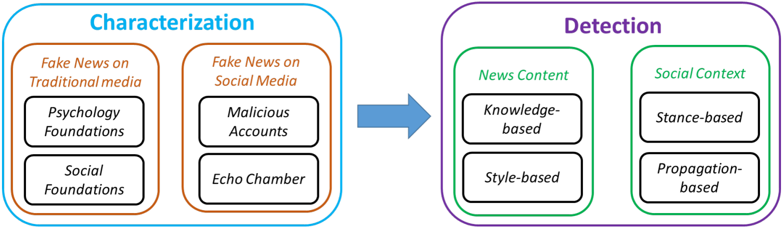
Framework



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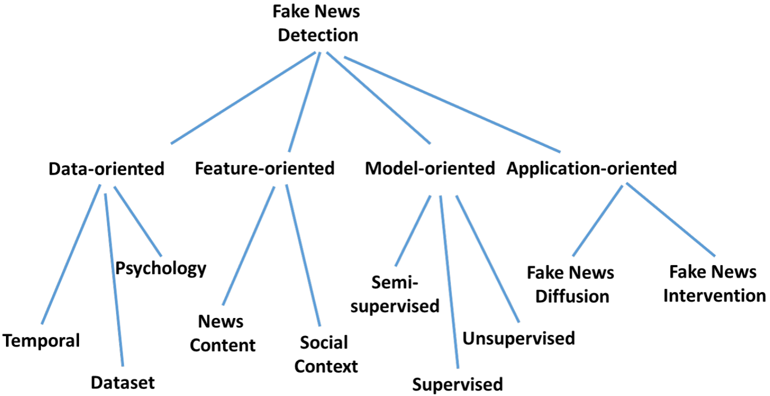
Future Directions



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Building a Bayesian Model for Events

Before reading anything,
we can use prior knowledge from historical data.

Valid Probability Model

A probability model must satisfy:

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Valid Probability Model

A probability model must satisfy:

1. Cover all possible events.
2. Assign probabilities to each event.
3. Probabilities sum to one.

We're Using R Instead of Python

Python

- "Can do everything" ... installs n packages, breaks after update
- Stack Overflow becomes your TA

R $\rightarrow$ `https://cran.r-project.org`

- Literally for statistics; `lm(y ~ x)` and move on with your life
- Excellent, open-source, keeps us focused on data analysis,
  ... and I enjoy sleeping at night!

What I Need From You: learn concepts, analyze data, pass the class!
What I Do Not Need: a 2-hour debate about virtual environments.

Dataset

```r
# Load packages
library(bayesrules)
library(tidyverse)
library(janitor)

# Import article data
data(fake_news)

fake_news %>%
  tabyl(type) %>%
  adorn_totals("row")
#  type   n percent
#  fake  60     0.4
#  real  90     0.6
# Total 150     1.0
```

Prior Probability Model

- 150 news articles:
  60 fake, 90 real
- $P(\text{Fake}) = \frac{60}{150} = 0.40$.
  Before looking at the headline, the prior probability that an article is fake is 40%.

Prior Belief: there is a 40% chance an article is fake before observing any evidence.

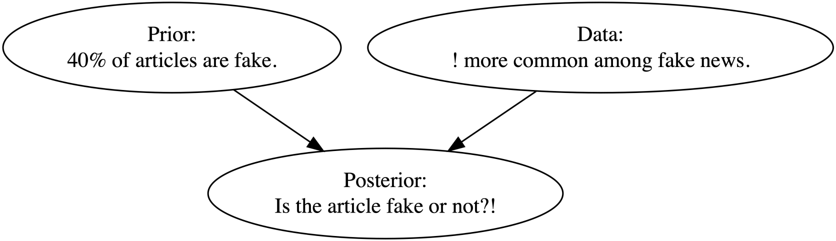
$$P(B) = 0.40 \quad \leftrightarrow \quad P(B^c) = 0.60$$

where B: article is **fake**, B^c: article is **real**.

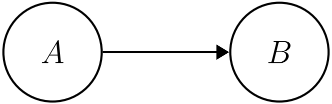
$$P(B) + P(B^c) = 0.40 + 0.60 = 1$$

Event	B	B^c	Total
Probability	0.4	0.6	1

Bayesian reasoning to estimate whether a news article is fake:



DAG: Directed Acyclic Graph



A = exclamation points "!" B = fake status "🥸"

- A **DAG** shows how variables are connected.
- The arrow means A may be associate with B.
- **Acyclic** means the arrows do not loop back.

DAGs and Conditional Probability

- The DAG suggests that the chance of B may depend on A.

$$P(B \mid A)$$

In words: the probability of article is fake **given**
the exclamation point in the title.

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In words: the probability of article is fake **given** the exclamation point in the title.

- If A changes the probability of B, then the events are **dependent**.

$$P(B \mid A) \neq P(B)$$

- If knowing A does not change the probability of B, then they are **independent**.

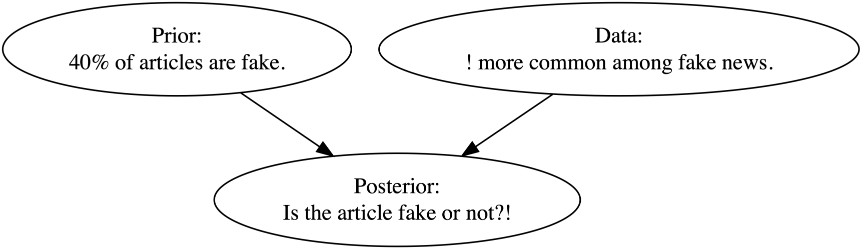
$$P(B \mid A) = P(B)$$

```r
# Transposed version of head()
# ... makes possible to see all columns in a data frame
glimpse(fake_news)
```

Observed headline pattern:

```r
# Tabulate exclamation usage and article type
fake_news %>%
  tabyl(title_has_excl, type) %>%
  adorn_totals("row")
# title_has_excl fake real
#          FALSE   44   88
#           TRUE   16    2
#          Total   60   90
```

Exclamation Points as Evidence



Title Has "!"	Fake	Real	Total
Yes	16	2	18
No	44	88	132
Total	60	90	150

Conditional Probability & Likelihood

Title Has "!"	Fake	Real	Total
Yes	16	2	18
No	44	88	132
Total	60	90	150

$$P(! \mid \text{Fake}) = \frac{16}{60} \approx 0.267,\ P(! \mid \text{Real}) = \frac{2}{90} \approx 0.022$$

An exclamation point increases the probability that an article is fake.

Conditional Probability & Likelihood

$$B : \text{article is fake}$$
$$B^c : \text{article is real}$$

$$A : \text{article uses exclamation points}$$
$$A^c : \text{article does not use exclamation points}$$

From the data:

$$P(A \mid B) = 0.267 \qquad P(A \mid B^c) = 0.022$$

Fake news articles are **much more likely** to use exclamation points than real news articles.

Conditional vs Unconditional Probability

- **Unconditional probability** is the chance something happens without extra information.

$$P(A)$$

- **Conditional probability** is the chance something happens given that something else is already known.

$$P(A \mid B)$$

- Comparing $P(A)$ and $P(A \mid B)$ tells us whether knowing B changes how likely A is.

Independent Events

- Two events are **independent** if knowing one happens does not change the probability of the other.
- *Example*: rolling a die and flipping a coin are independent, because the die result **does not** affect the coin flip.
- In symbols:

$$P(A \mid B) = P(A)$$

From Probability to Likelihood

- Earlier, we computed:

$$P(A \mid B) = 0.2667 \qquad P(A \mid B^c) = 0.0222$$

- Fake articles are much more likely to use exclamation points.
- Suppose we already observed an article with exclamations.

  Now the question becomes:

  Is the article more likely fake or real?

What is Likelihood?

- **Probability** starts with a known cause and predicts data.

$$P(A \mid B)$$

- **Likelihood** starts with observed data and compares possible causes.

$$\mathcal{L}(B \mid A) = P(A \mid B)$$

- In our example:
  - A: article uses exclamation points
  - B: article is fake
  - Since

  $$P(A \mid B) > P(A \mid B^c)$$

  exclamation provide evidence that the article may be fake.

Probability	**Likelihood**
Known condition	Known data
Predicts outcomes	Compares explanations
$P(A \mid B)$	$\mathcal{L}(B \mid A)$

- The distinction is **subtle**, especially since people use the terms "likelihood" and "probability" interchangeably in casual conversation.
- Likelihood **helps us decide** which explanation best fits the observed data.

- The likelihood function **is not** a probability function, but rather provides a framework to compare the relative compatibility of our exclamation point data with B and B^c.

Event	B	B^c	**Total**
Prior probability	0.4	0.6	1
Likelihood $(\cdot \mid A)$	0.2667	0.0232	0.2889

Code: https://abuabara.github.io/DAEN427F26/Lecture2.R